Project Report: Portfolio Generator Web App

Introduction

. This project focuses on building a **Next.js web application** that allows users to upload Markdown (`.md`) files and automatically convert them into a styled portfolio website. The goal was to create a reproducible, demo-ready solution that can be hosted on Vercel for easy sharing via a live URL

Abstract

The Portfolio Generator simplifies the process of showcasing personal or professional content. By uploading a Markdown file, users can instantly generate a portfolio site with selectable themes. The app leverages modern web technologies to ensure clarity, accessibility, and reproducibility. Deployment on Vercel makes the portfolio publicly accessible, enabling reviewers or employers to view the content without requiring local setup.

Tools Used

- **Next.js** – React framework for building the web app
- **React** – UI library for interactive components
- **Tailwind CSS** – Styling and theme management
- **Node.js & npm** – Package management and local development
- **Vercel** – Hosting and deployment platform

Steps Involved in Building the Project

1. **Project Setup**: Initialized a Next.js app using `create-next-app`.
2. **File Structure**: Created essential files (`pages/index.js`, `components/MarkdownRenderer.js`, `styles/globals.css`, etc.).
3. **Markdown Upload & Parsing**: Implemented file upload functionality and parsing logic to convert `.md` into HTML.
4. **Theme Integration**: Added multiple themes (light, dark, modern) with a theme switcher.
5. **Styling**: Applied Tailwind CSS for responsive and clean design.
6. **Testing Locally**: Ran the app on `http://localhost:3000` to verify functionality.
7. **Deployment**: Authorized and deployed the project on Vercel using `npx vercel`, generating a live shareable URL.

Conclusion

The Portfolio Generator successfully demonstrates how Markdown content can be transformed into a polished, themed portfolio site. The project highlights reproducibility, automation, and accessibility, aligning with professional review standards. Hosting on Vercel ensures that the portfolio is easily shareable and ready for real-world use.